# Report

## Hyperparameters

### MoE

```
TRAIN_TOKEN_CHOICE = True          # Train Token Choice router
TRAIN_HASH = False               # Train Hash router
TRAIN_WITH_LOAD_BALANCER = True    # Train with load balancer
TRAIN_WITHOUT_LOAD_BALANCER = False # Train without load balancer
USE_CUSTOM_GQA = False              # Use custom GQA (Bonus 2)
USE_LORA = True                     # Use LoRA for experts (Bonus 3)

# Model Architecture
D_MODEL = 512
NHEAD = 8
NUM_ENCODER_LAYERS = 6
NUM_DECODER_LAYERS = 6
NUM_EXPERTS = 4
D_FF = 1024
TOP_K = 2
DROPOUT = 0.1
MAX_SEQ_LEN = 512
GQA_NUM_KV_HEADS = 4  # For GQA (must divide NHEAD)

# LoRA Configuration (Bonus 3)
LORA_RANK = 16                      # LoRA rank (lower = more compression)
LORA_ALPHA = 32                     # LoRA scaling factor
LORA_DROPOUT = 0.1                  # LoRA dropout

# Load Balancing
LOAD_BALANCE_ALPHA = 0.01

# Training Settings
BATCH_SIZE = 32
NUM_EPOCHS = 1
LEARNING_RATE = 5e-5
```

```
    WARMUP_STEPS = 500
    GRAD_CLIP = 1.0
    MAX_LENGTH = 256

    # Dataset
    DATASET_NAME = "EdinburghNLP/xsum"
    TRAIN_SAMPLES = 100000  # Set to None for full dataset
    VAL_SAMPLES = 2000
    TEST_SAMPLES = 2000

    # Tokenizer
    TOKENIZER_NAME = "t5-small"

    # Checkpointing
    CHECKPOINT_DIR = "./checkpoints"
    SAVE_EVERY_N_EPOCHS = 1

    # Hugging Face
    HF_USERNAME = "J10Official"
    PUSH_TO_HUB = True

    # Evaluation
    EVAL_SAMPLES = 500
    GENERATION_MAX_LENGTH = 100
```

## T5

```
training_args = Seq2SeqTrainingArguments(
    output_dir="./flan_t5_xsum_lora",  # Changed to reflect LoRA training
    eval_strategy="steps",  # Changed from evaluation_strategy in latest transformers
    eval_steps=1000,  # Less frequent eval (was 500)
    learning_rate=3e-4,  # Higher learning rate for LoRA (was 5e-5)
    per_device_train_batch_size=8,  # Balanced for T4 GPU speed
    per_device_eval_batch_size=8,   # Balanced for T4 GPU
    gradient_accumulation_steps=1,  # NO accumulation - direct updates
    num_train_epochs=3,  # Train for 3 epochs
    weight_decay=0.01,
    save_total_limit=2,  # Save fewer checkpoints
```

```
    save_steps=125,  # Checkpoint every 1000 samples (1000 samples / 8 batch_size = 125 steps)
    logging_steps=50,  # More frequent logging to see progress
    predict_with_generate=True,
    generation_max_length=64,
    fp16=True,  # Mixed precision training
    gradient_checkpointing=False,  # DISABLE - it slows down training!
    push_to_hub=False,
    load_best_model_at_end=True,
    metric_for_best_model="rouge1",
    greater_is_better=True,
    report_to="none",
    warmup_steps=500,
    dataloader_num_workers=2,  # Parallel data loading
    dataloader_pin_memory=True,  # Faster data transfer to GPU
)
```

## Llama

```
Llama-1B    unsloth/Llama-3.2-1B    QLoRA (Unsloth) 2e-4    16  16  4-bit quantization.
```

### Performance Summary

| Model | ROUGE-1 | ROUGE-2 | ROUGE-L | BLEU | BERTScore F1 |
|---|---|---|---|---|---|
| BART-Base | 45.17 | 21.80 | 36.64 | | |
| FLAN-T5-Base-LoRA | 36.70 | 13.73 | 28.84 | | |
| Llama-1B-LoRA | 23.41 | 7.39 | 17.68 | 3.01 | 87.26 |
| sparse-moe-Token-Choice | 32.22 | 10.30 | 24.86 | 38.50 | 88.78 |
| sparse-moe-Hash | 29.40 | 8.43 | 22.58 | 37.20 | 87.98 |

### Document Metrics

| Model | Compression Ratio | Extractiveness (%) |
|---|---|---|
| BART-Base | 0.0920 | 66.40 |

| Model | Compression Ratio | Extractiveness (%) |
|---|---|---|
| FLAN-T5-Base-LoRA | 0.0915 | 65.20 |
| Llama-1B-LoRA | 0.27 | 75.78 |
| sparse-moe-Token-Choice | 0.0894 | 60.87 |
| sparse-moe-Hash | 0.0882 | 53.66 |

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- Hugging Face: https://huggingface.co/J10Official
- GitHub: https://github.com/ANLP2025/assignment3-J10Official